

MSc DSAI 20242025 AIEDA323 Biomedical Signal Processing Computer Labs (1) - Professor Vicente Zarzoso

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1 Spectral analysis and filtering

1.1 1.1 Abdominal ECG during pregnancy

The provided data file contains an 8-lead ECG recorded from a pregnant woman at a 500-Hz sampling rate. The first five leads were obtained from the mothers abdominal region, where as the last three leads were recorded from her thoracic region.

1. Plot the 8 leads. Remark that the 4th abdominal lead is particularly noisy.

```
% Load the data
load('C:\Users\praba\Documents\GitHub\UCA-M2-SEMESTER2\bio medical signal processing\lab
    assignments\ecg_mother.mat');
Fs = 500;
N = 5000;
t = (0:N-1)/Fs;

% 1. Plot the 8 leads
figure;
for i = 1:8
    subplot(8,1,i);
    plot(t, Y(i,:));
    ylabel(['Lead ', num2str(i)]);
    if i == 1
        title('8-Lead ECG Recording (Leads 1-5: Abdominal, 6-8: Thoracic)');
    end
end
xlabel('Time (s)');
```

- The 4th abdominal lead (Lead 4) appears particularly noisy compared to the other leads. This could be due to maternal muscle activity or other interference during the recording process.
- The noisy abdominal lead (Lead 4) may require pre-processing, such as filtering or artifact removal, to improve its quality for further analysis.

2. Using the spectral estimation method of your choice, compute and plot the power spectral density (PSD) of the noisy lead.

```
fs = 500;
noisy_lead = Y(4,:); % 4th abdominal lead

% Welch's method parameters
segmentLength = 512; % Segment length (you can tune this)
window = hamming(segmentLength); % Hamming window
```

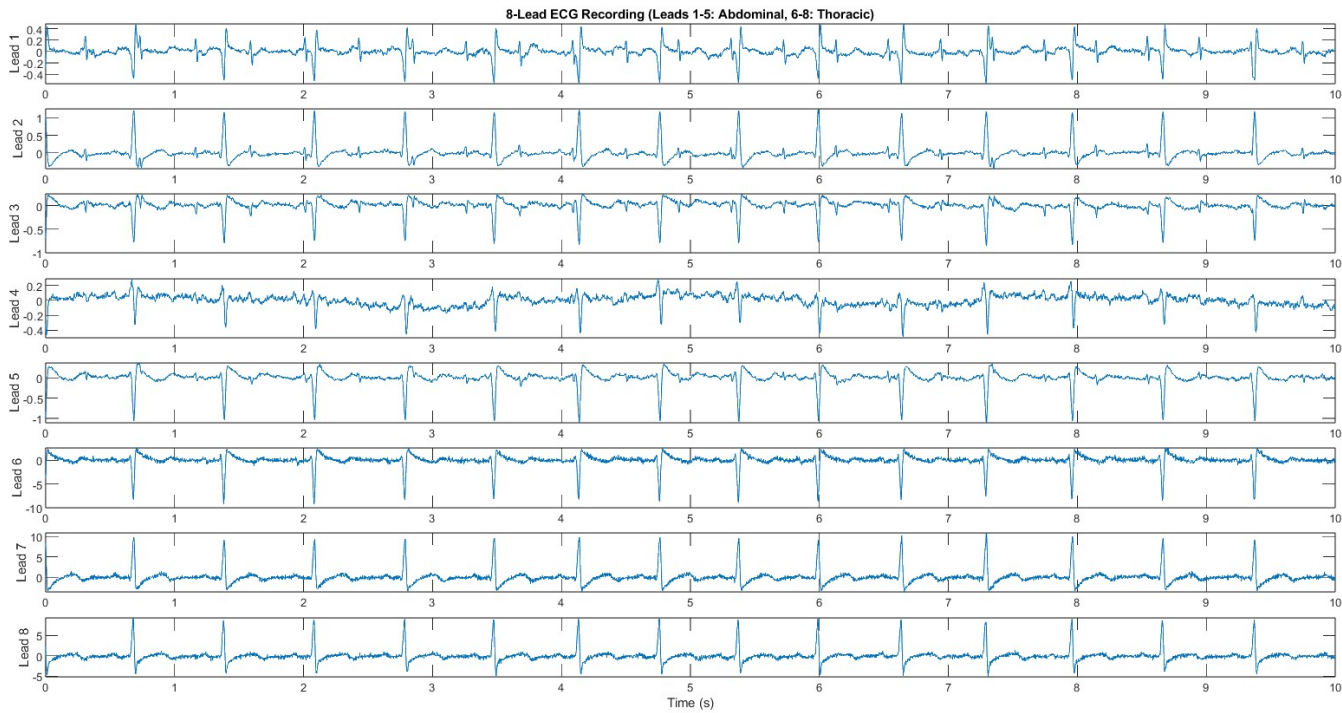


Fig. 1: 8 leads.

```

overlap = segmentLength / 2;          % 50% overlap
nfft = 2048;                          % Number of FFT points for better resolution

% Estimate PSD using Welch's method
[pxx, f] = pwelch(noisy_lead, window, overlap, nfft, fs);

% Plot PSD
figure;
plot(f, 10*log10(pxx), 'b', 'LineWidth', 1.5);
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
title('Welch Power Spectral Density - Noisy Abdominal ECG (Lead 4)');
grid on;
xlim([0 100]); % Focus on ECG + noise band

% Highlight known noise regions
hold on;
yL = ylim;

% Power line interference at 50 Hz
line([50 50], yL, 'Color', 'r', 'LineStyle', '--');
text(52, yL(2)-10, '50 Hz (Power Line)', 'Color', 'r');

% Baseline wander (<1 Hz)
area(f(f < 1), 10*log10(pxx(f < 1)), ...
    'FaceColor', [0.8 0.8 1], 'EdgeColor', 'none', 'FaceAlpha', 0.3);
legend('PSD (Welch)', 'Power Line Noise', 'Baseline Wander Region');

```

- The Power Spectral Density (PSD) graph represents the distribution of power across different frequency components of the noisy abdominal ECG signal (Lead 4). It is computed using Welch's method, which provides a smoothed estimate of the PSD by dividing the signal into overlapping segments, applying a window function, and averaging the periodograms.
- The negative PSD values are a result of the logarithmic scale and the low power of the signal,

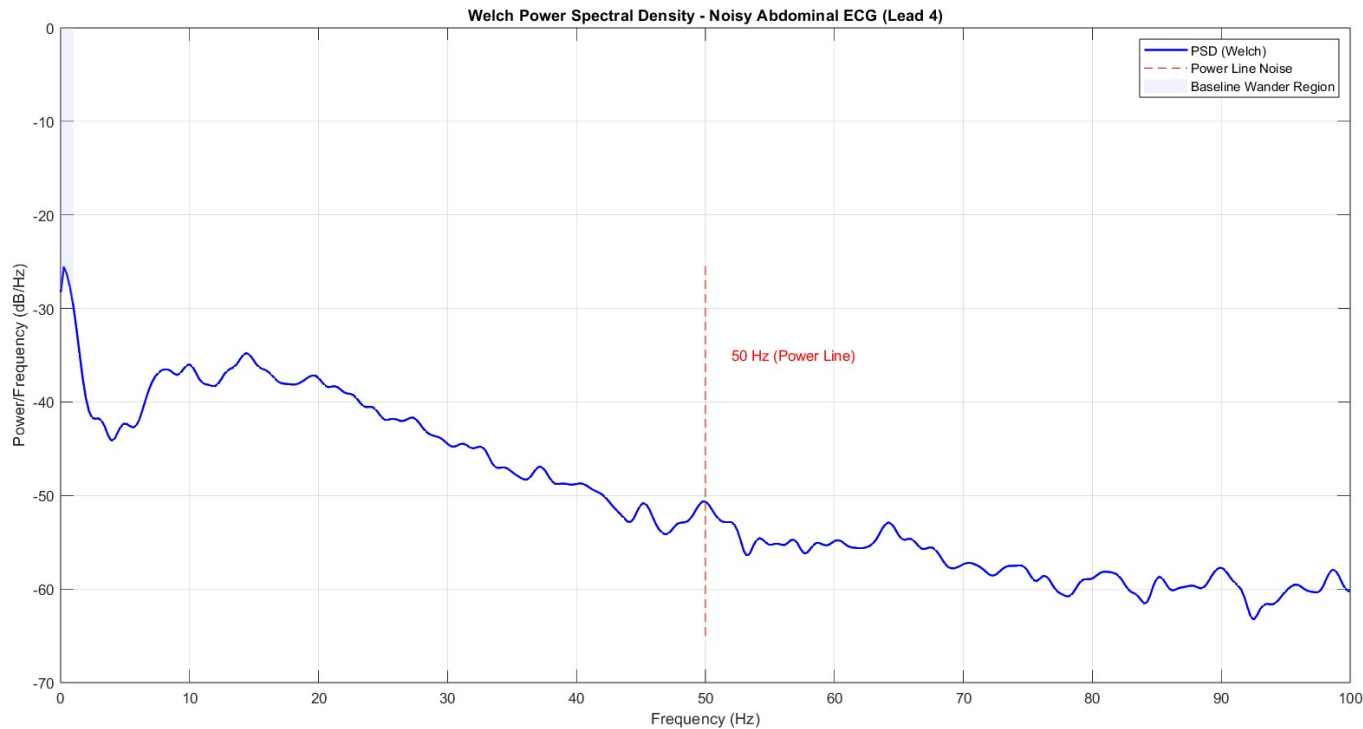


Fig. 2: PSD for Lead 4 (noisy lead)

3. By observing the estimated PSD, identify the frequency intervals of noise and interference contaminating the signal of interest.

- **Baseline Wander (Low-Frequency Noise):**
 - **Frequency Range:** Below 1 Hz.
 - **Cause:** Caused by slow variations such as respiration, patient movement, or electrode drift.
 - **Observation:** The PSD shows a high power in this region, which is typical for baseline wander.
- **Power Line Interference:**
 - **Frequency Range:** Around 50 Hz (marked by the red dashed line).
 - **Cause:** Caused by electrical noise from the power supply.
 - **Observation:** A sharp peak is visible at 50 Hz, indicating significant power line interference.
- **High-Frequency Noise:**
 - **Frequency Range:** Above 40 Hz.
 - **Cause:** Likely due to muscle activity (electromyographic artifacts) or electronic noise from the recording equipment.
 - **Observation:** The PSD shows small peaks and a gradual decline in power in this range.

4. Design appropriate frequency filters to reduce noise and interference.

- **High-Pass Filter for Baseline Wander Removal (less than 1 Hz)**
 - **Purpose:** Remove low-frequency baseline wander caused by respiration or patient movement.
 - **Filter Design:**
 - * Use a high-pass filter with a cutoff frequency of **1 Hz**.
 - * Filter type: Butterworth

```
% High-pass filter design
hp_cutoff = 1; % Cutoff frequency (Hz)
[b_hp, a_hp] = butter(4, hp_cutoff / (fs / 2), 'high'); % 4th-order Butterworth filter

% Apply the high-pass filter
filtered_signal_hp = filtfilt(b_hp, a_hp, noisy_lead);
```

Listing 1.1: High-pass filter for baseline wander removal

– Notch Filter for Power Line Interference (50 Hz)

- **Purpose:** Remove the sharp peak at **50 Hz** caused by power line interference.
- **Filter Design:**
 - * Use a notch filter centered at **50 Hz** with a narrow bandwidth.
 - * Filter type: IIR (Butterworth).

```
% Notch filter design
notch_freq = 50; % Notch frequency (Hz)
notch_bw = 1; % Bandwidth (Hz)
wo = notch_freq / (fs / 2); % Normalized frequency
bw = notch_bw / (fs / 2); % Normalized bandwidth

% Design the notch filter
[b_notch, a_notch] = butter(2, [wo - bw/2, wo + bw/2], 'stop'); % 2nd-order Butterworth
bandstop filter

% Apply the notch filter
filtered_signal_notch = filtfilt(b_notch, a_notch, filtered_signal_hp);
```

Listing 1.2: Notch filter for 50 Hz power line interference

– Low-Pass Filter for High-Frequency Noise (greater than 40 Hz)

- **Purpose:** Remove high-frequency noise caused by muscle activity or electronic artifacts.
- **Filter Design:**
 - * Use a low-pass filter with a cutoff frequency of **40 Hz**.
 - * Filter type: Butterworth

```
% Low-pass filter design
lp_cutoff = 40; % Cutoff frequency (Hz)
[b_lp, a_lp] = butter(4, lp_cutoff / (fs / 2), 'low'); % 4th-order Butterworth filter

% Apply the low-pass filter
filtered_signal = filtfilt(b_lp, a_lp, filtered_signal_notch);
```

Listing 1.3: Low-pass filter for high-frequency noise

– Combined Filtering

- Apply filters in sequence:
 1. High-pass filter to remove baseline wander.
 2. Notch filter to remove power line interference.
 3. Low-pass filter to remove high-frequency noise.

```
% Apply all filters sequentially
filtered_signal_hp = filtfilt(b_hp, a_hp, noisy_lead); % High-pass filter
filtered_signal_notch = filtfilt(b_notch, a_notch, filtered_signal_hp); % Notch filter
filtered_signal = filtfilt(b_lp, a_lp, filtered_signal_notch); % Low-pass filter

% Plot the filtered signal
figure;
plot(t, filtered_signal);
title('Filtered ECG Signal (Lead 4)');
xlabel('Time (s)');
ylabel('Amplitude (mV)');
```

Listing 1.4: Applying all filters sequentially

– Expected Outcome

- **Baseline wander (≥ 1 Hz)** will be removed by the high-pass filter.
- **Power line interference (50 Hz)** will be eliminated by the notch filter.
- **High-frequency noise (≥ 40 Hz)** will be reduced by the low-pass filter.
- The resulting signal will retain the ECG components (P-wave, QRS complex, T-wave) in the **1–40 Hz** range. In the filtered signal noise is reduced.

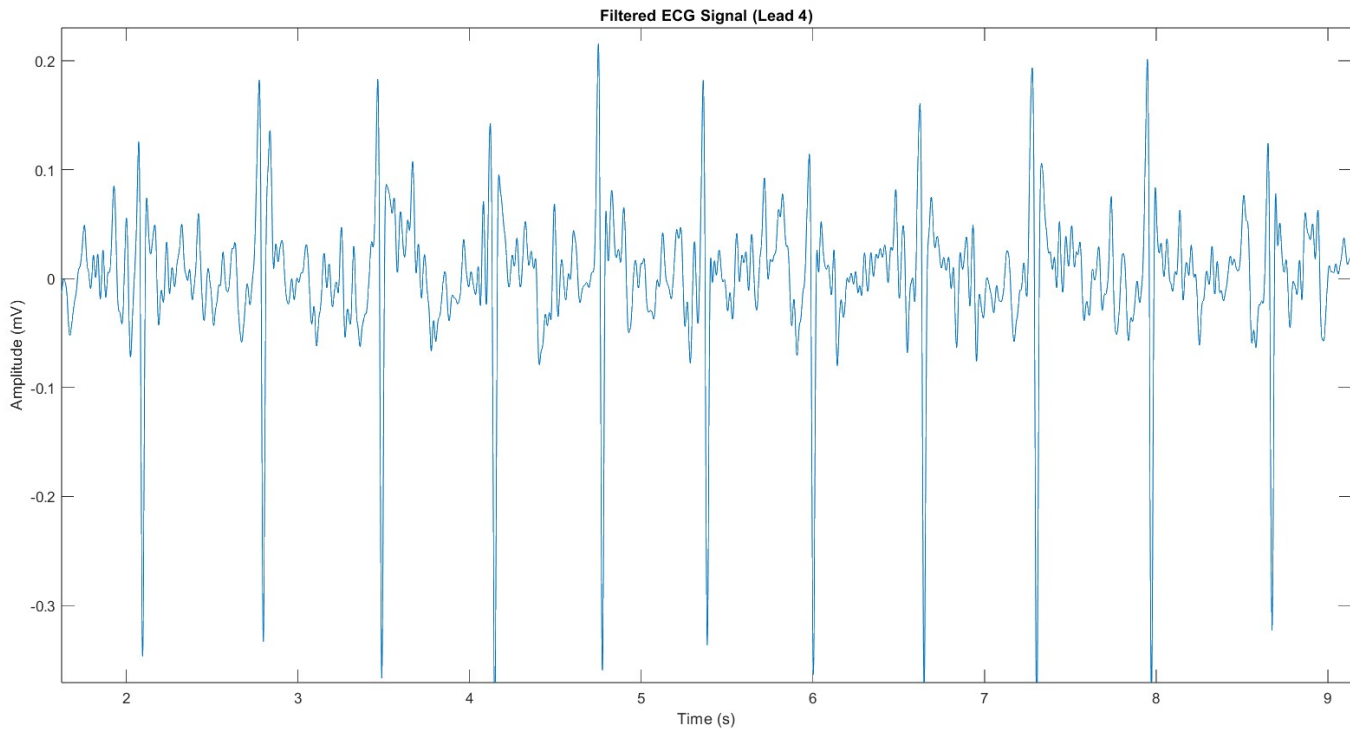


Fig. 3: Filtered ECG Signal(lead 4)

5. Apply the filters to the observed ECG and confirm their effectiveness by analyzing the output signal in both time and frequency domains To apply the filters to the observed ECG signal and confirm their effectiveness, we analyze the output signal in both the **time domain** (to observe noise reduction) and the **frequency domain** (to confirm the removal of specific noise components). The following steps detail the procedure.

– **Apply Filters to the ECG Signal**

- The filters include:
 - * **High-pass filter:** Removes baseline wander (<1 Hz).
 - * **Notch filter:** Removes power line interference (50 Hz).
 - * **Low-pass filter:** Removes high-frequency noise (>40 Hz).
- Filters are applied sequentially as shown below:

```
% Apply all filters sequentially
filtered_signal_hp = filtfilt(b_hp, a_hp, noisy_lead); % High-pass filter
filtered_signal_notch = filtfilt(b_notch, a_notch, filtered_signal_hp); % Notch filter
filtered_signal = filtfilt(b_lp, a_lp, filtered_signal_notch); % Low-pass filter
```

Listing 1.5: Sequential application of filters

– **Analyze the Filtered Signal in the Time Domain**

- Plot the original and filtered signals to visually assess the reduction in noise.

```
% Plot original and filtered signals in the time domain
figure;
subplot(2, 1, 1);
plot(t, noisy_lead, 'r');
title('Original Noisy ECG Signal (Lead 4)');
xlabel('Time (s)');
ylabel('Amplitude (mV)');
grid on;

subplot(2, 1, 2);
plot(t, filtered_signal, 'b');
title('Filtered ECG Signal (Lead 4)');
xlabel('Time (s)');
```

```
ylabel('Amplitude (mV)');
grid on;
```

Listing 1.6: Time domain visualization of original and filtered signals

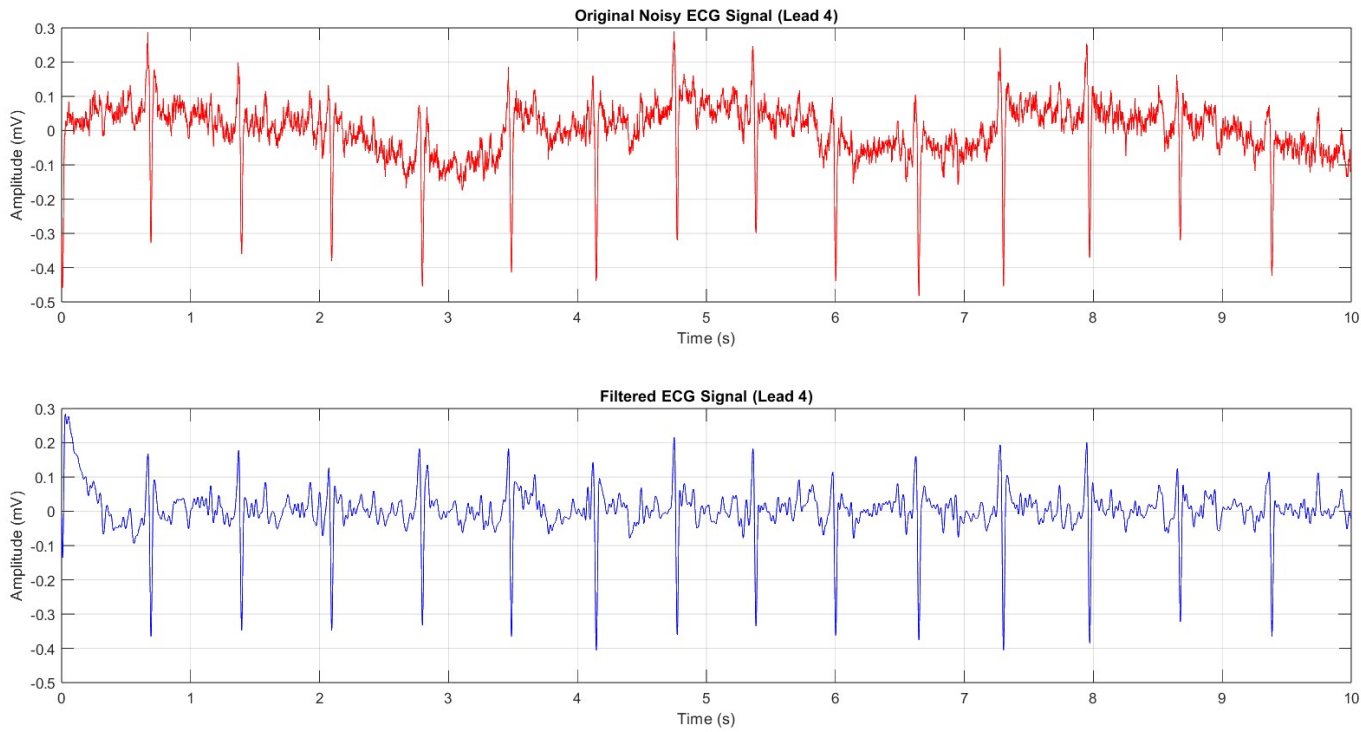


Fig. 4: Filtered ECG Signal(lead 4)

– Analyze the Filtered Signal in the Frequency Domain

- Use Welch's method to compute the Power Spectral Density (PSD) and compare the spectral content of the original and filtered signals.

```
% Compute PSD of original and filtered signals
[pxx_original, f] = pwelch(noisy_lead, window, overlap, nfft, fs);
[pxx_filtered, f] = pwelch(filtered_signal, window, overlap, nfft, fs);

% Plot PSDs
figure;
plot(f, 10*log10(pxx_original), 'r', 'LineWidth', 1.5); hold on;
plot(f, 10*log10(pxx_filtered), 'b', 'LineWidth', 1.5);
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
title('PSD of Original and Filtered ECG Signals');
legend('Original Signal', 'Filtered Signal');
grid on;
xlim([0 100]); % Focus on the relevant frequency range
```

Listing 1.7: Frequency domain analysis using PSD

– 4. Expected Observations

- **Time Domain:**
 - * The filtered signal show a smoother waveform with reduced baseline wander, power line interference, and high-frequency noise.
 - * ECG components like the QRS complex, P-wave, and T-wave remain intact.
- **Frequency Domain:**
 - * The baseline wander component (<1 Hz) should be significantly reduced.

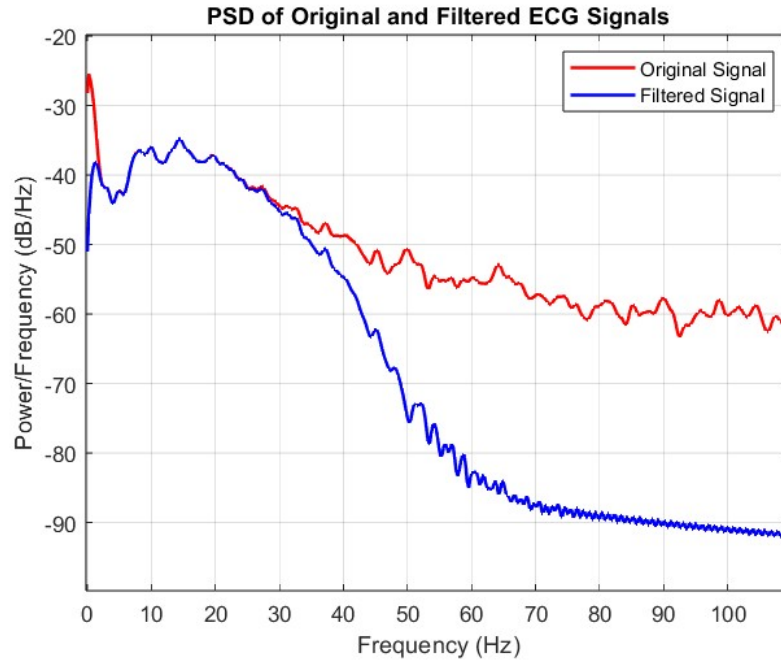


Fig. 5: PSD of original and Filtered ECG Signal

- * The sharp 50 Hz power line interference peak removed.
- * High-frequency noise (>40 Hz) attenuated, preserving the 1–40 Hz band.
- By comparing the time-domain and frequency-domain plots, we confirm that the applied filtering approach effectively reduces noise and interference while preserving clinically relevant ECG features.

1.2 Atrial Fibrillation Analysis [80%]

Spectral parameters of the atrial fibrillation (AF) ECG have been used as markers of noninvasive atrial activity estimation performance and predictors of therapy outcome. Two such parameters are:

- **Dominant Frequency (DF)**: Defined as the PSD peak position, f_p , in the interval [3–9] Hz, the typical AF band. In the literature, low values of this parameter have been linked to high probabilities of spontaneous cardioversion and favorable therapy outcome prediction.
- **Spectral Concentration (SC)**: Defined as the percentage of signal power localized in a narrow frequency band around f_p . It quantifies the harmonic character or repetitiveness of the atrial activity signal. High values of this parameter have been associated with clean atrial signals, weakly contaminated by residual ventricular components.

This experiment aims at assessing the ability of these spectral parameters to characterize the AF data provided in the module.

MATLAB functions to compute the DF and SC of the signals given as inputs. Use the Welch method as the PSD estimator with typical parameters from the literature: window type and size, overlap factor, and number of FFT points. Also identify the classical frequency band for SC computation from the literature.

```
function [DF, SC] = compute_df_sc(signal, fs)
% Function to compute Dominant Frequency (DF) and Spectral Concentration (SC)
% Inputs:
%   signal - Input AF signal
%   fs - Sampling frequency (Hz)
% Outputs:
%   DF - Dominant Frequency (Hz)
%   SC - Spectral Concentration (%)

% Parameters for Welch's method
```

```

segmentLength = 512;           % Segment length
window = hamming(segmentLength); % Hamming window
overlap = segmentLength / 2;   % 50% overlap
nfft = 2048;                   % Number of FFT points

% Compute PSD using Welch's method
[pxx, f] = pwelch(signal, window, overlap, nfft, fs);

% Define the AF band for analysis (3-9 Hz)
af_band = [3, 9]; % AF band in Hz
af_band_idx = (f >= af_band(1)) & (f <= af_band(2)); % Indices for AF band
f_af = f(af_band_idx); % Frequencies in AF band
pxx_af = pxx(af_band_idx); % PSD values in AF band

% Compute Dominant Frequency (DF)
[~, max_idx] = max(pxx_af); % Find the peak in the AF band
DF = f_af(max_idx); % Dominant frequency

% Compute Spectral Concentration (SC)
narrow_band_width = 0.5; % Narrow band width around DF (0.5 Hz)
narrow_band_idx = (f_af >= (DF - narrow_band_width)) & (f_af <= (DF + narrow_band_width));
narrow_band_power = sum(pxx_af(narrow_band_idx)); % Power in narrow band
total_af_power = sum(pxx_af); % Total power in AF band
SC = (narrow_band_power / total_af_power) * 100; % Spectral concentration as a percentage
end

```

– **Welch's Method:**

- The Power Spectral Density (PSD) is estimated using Welch's method.
- Parameters used: Hamming window, 50% overlap, and 2048 FFT points for improved frequency resolution.

– **Dominant Frequency (DF):**

- The Dominant Frequency is identified as the frequency corresponding to the peak of the PSD within the atrial fibrillation (AF) band (3–9 Hz).

– **Spectral Concentration (SC):**

- SC is calculated as the percentage of signal power localized within a narrow band of ± 0.5 Hz around the DF.
- This value is expressed relative to the total power in the AF band (3–9 Hz).

1.2 Compute the DF and SC in lead V1 of all ECG records in the provided AF dataset.

```

% Parameters
fs = 256; % Sampling frequency
lead = 7; % Lead V1 (7th lead in the dataset)
num_patients = size(Xa_all, 3); % Total number of patients

% Initialize results
DF_results = zeros(num_patients, 1); % Dominant Frequency for each patient
SC_results = zeros(num_patients, 1); % Spectral Concentration for each patient

% Loop through all patients
for patient = 1:num_patients
    % Extract the signal for lead V1 of the current patient
    signal = squeeze(Xa_all(lead, :, patient));

    % Compute DF and SC using the provided function
    [DF, SC] = compute_df_sc(signal, fs);

    % Store results
    DF_results(patient) = DF;
    SC_results(patient) = SC;
end

% Display summary results
fprintf('Mean Dominant Frequency (DF): %.2f Hz\n', mean(DF_results));
fprintf('Mean Spectral Concentration (SC): %.2f %%\n', mean(SC_results));

```

```
% Plot DF and SC distributions
figure;
subplot(2, 1, 1);
histogram(DF_results, 10, 'FaceColor', 'b');
title('Dominant Frequency (DF) Distribution');
xlabel('Frequency (Hz)'); ylabel('Count');

subplot(2, 1, 2);
histogram(SC_results, 10, 'FaceColor', 'r');
title('Spectral Concentration (SC) Distribution');
xlabel('SC (%)'); ylabel('Count');
```

```
>> *****output result*****
```

```
Mean Dominant Frequency (DF): 6.71 Hz
```

```
Mean Spectral Concentration (SC): 53.13 %
```

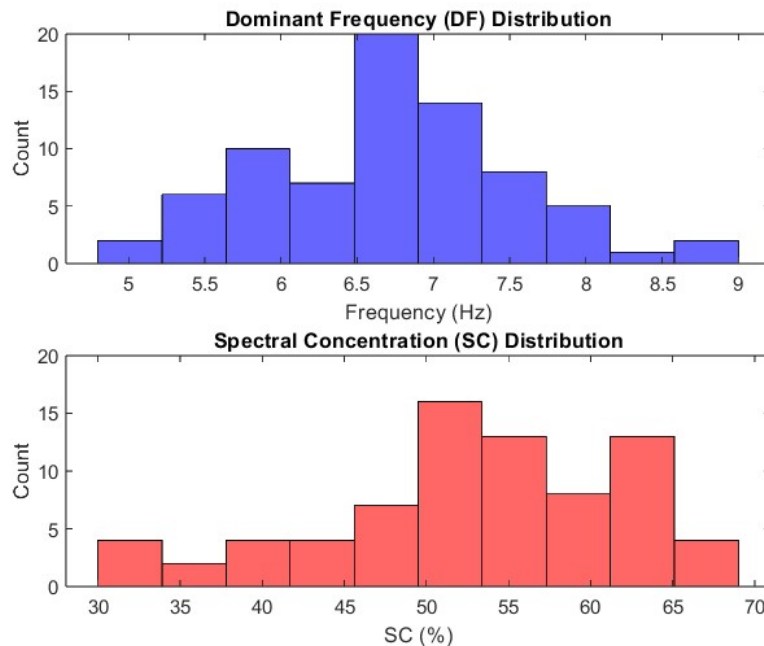


Fig. 6: PSD of original and Filtered ECG Signal

– **Mean DF and SC:**

- The average Dominant Frequency (DF) and Spectral Concentration (SC) values across all patients are displayed in the console.

– **Dominant Frequency (DF) Distribution (Top Graph):**

- **X-axis:** Frequency (Hz) in the AF band (3–9 Hz).
- **Y-axis:** Count of patients with a specific DF value.
- **Observation:**
 - * DF values are concentrated around **6.5–7 Hz**.
 - * Few patients have DF values lower or higher than this range.

– **Spectral Concentration (SC) Distribution (Bottom Graph):**

- **X-axis:** Spectral Concentration (SC) in percentage (%).
- **Y-axis:** Count of patients with a specific SC value.
- **Observation:**
 - * SC values range between **30% and 70%**, peaking around **50%**.

1.3 Using box-and-whiskers plots, compare the distributions of the spectral parameters of the recorded data (X_{va}) versus those of the estimated atrial activity (X_a) over the patient population.

```

% Parameters
fs = 256; % Sampling frequency
lead = 7; % Lead V1 (7th lead in the dataset)
num_patients = size(Xva_all, 3); % Total number of patients

% Initialize results
DF_Xva = zeros(num_patients, 1); % Dominant Frequency for Xva
SC_Xva = zeros(num_patients, 1); % Spectral Concentration for Xva
DF_Xa = zeros(num_patients, 1); % Dominant Frequency for Xa
SC_Xa = zeros(num_patients, 1); % Spectral Concentration for Xa

% Loop through all patients
for patient = 1:num_patients
    % Extract the signal for lead V1 of the current patient
    signal_Xva = squeeze(Xva_all(lead, :, patient)); % Raw ECG (Xva)
    signal_Xa = squeeze(Xa_all(lead, :, patient)); % Atrial activity (Xa)

    % Compute DF and SC for Xva
    [DF_Xva(patient), SC_Xva(patient)] = compute_df_sc(signal_Xva, fs);

    % Compute DF and SC for Xa
    [DF_Xa(patient), SC_Xa(patient)] = compute_df_sc(signal_Xa, fs);
end

% Combine data for box-and-whiskers plots
DF_data = [DF_Xva; DF_Xa];
SC_data = [SC_Xva; SC_Xa];
group_labels = categorical([repmat("Xva (Raw ECG)", num_patients, 1); repmat("Xa (Atrial
    Activity)", num_patients, 1)]);

% Create box-and-whiskers plots
figure;

% Dominant Frequency (DF) comparison
subplot(2, 1, 1);
boxchart(group_labels, DF_data, 'BoxFaceColor', 'b');
title('Comparison of Dominant Frequency (DF)');
ylabel('Frequency (Hz)');

% Spectral Concentration (SC) comparison
subplot(2, 1, 2);
boxchart(group_labels, SC_data, 'BoxFaceColor', 'r');
title('Comparison of Spectral Concentration (SC)');
ylabel('Spectral Concentration (%)');

```

– Dominant Frequency (DF) Comparison (Top Plot):

- **X-axis:** Groups (Xva for raw ECG and Xa for atrial activity).
- **Y-axis:** Dominant Frequency (DF) in Hz.
- **Observations:**
 - * DF values for **Xa (Atrial Activity)** are more concentrated with a narrower interquartile range (IQR) and fewer outliers.
 - * This indicates that the atrial activity extraction process has effectively isolated the dominant atrial frequency within the AF band (3–9 Hz).
 - * DF values for **Xva (Raw ECG)** are more spread out, showing a wider IQR and several outliers.
 - * This spread is due to the presence of ventricular components and noise affecting the spectral analysis in raw ECG signals.
- **Interpretation:**
 - * The narrower DF distribution in Xa suggests effective isolation of atrial signals.
 - * DF becomes a more reliable marker for atrial activity in Xa compared to Xva.

– Spectral Concentration (SC) Comparison (Bottom Plot):

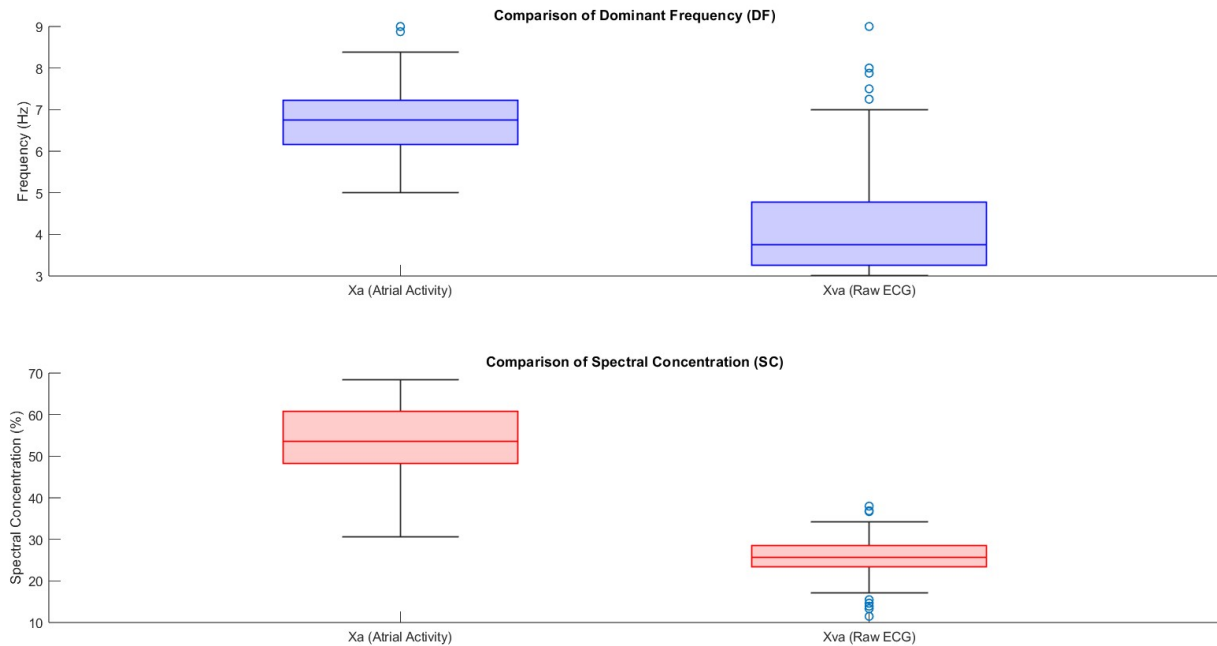


Fig. 7: PSD of original and Filtered ECG Signal

- **X-axis:** Groups (Xva for raw ECG and Xa for atrial activity).
- **Y-axis:** Spectral Concentration (SC) in percentage (%).
- **Observations:**
 - * SC values for **Xa (Atrial Activity)** are higher and more consistent, with a narrower IQR and fewer outliers.
 - * This reflects a higher concentration of power around the dominant frequency, indicating cleaner and more repetitive atrial signals.
 - * SC values for **Xva (Raw ECG)** are lower and more variable, showing a wider IQR and more outliers.
 - * This variability is due to noise and ventricular contamination in raw ECG signals, which reduces spectral concentration.
- **Interpretation:**
 - * Higher SC values in Xa suggest effective noise and ventricular component reduction.
 - * The atrial signals in Xa exhibit a more harmonic and cleaner structure.

The box-and-whiskers plots demonstrate that the spectral parameters Dominant Frequency (DF) and Spectral Concentration (SC) are more reliable and consistent in the estimated atrial activity signals (Xa) compared to the raw ECG signals (Xva). This highlights the effectiveness of the atrial activity extraction process in isolating meaningful atrial signals for spectral analysis.

1.4 Design a classifier to predict AF recurrence using DF and SC as features. Evaluate its performance using 5-fold cross-validation.

```
%formance using 5-fold cross-validation.

% Parameters
fs = 256; % Sampling frequency
lead = 7; % Lead V1 (7th lead in the dataset)
num_patients = size(Xva_all, 3); % Total number of patients

% Initialize results
DF_Xa = zeros(num_patients, 1); % Dominant Frequency for Xa
SC_Xa = zeros(num_patients, 1); % Spectral Concentration for Xa

% Extract recurrence labels
load('C:\Users\praba\Documents\GitHub\UCA-M2-SEMESTER2\bio medical signal processing\lab
assignments\Atrial_prabal_ghosh\indrecur.mat');
```

```

load('C:\Users\praba\Documents\GitHub\UCA-M2-SEMESTER2\bio medical signal processing\lab
    assignments\Atrial_prabal_ghosh\indnonrecur.mat');
y = zeros(num_patients, 1); % Initialize labels
y(indrecur) = 1; % Recurrence patients labeled as 1
y(indnonrecur) = 0; % Non-recurrence patients labeled as 0

% Loop through all patients to compute DF and SC for Xa
for patient = 1:num_patients
    % Extract the signal for lead V1 of the current patient
    signal_Xa = squeeze(Xa_all(lead, :, patient)); % Atrial activity (Xa)

    % Compute DF and SC for Xa
    [DF_Xa(patient), SC_Xa(patient)] = compute_df_sc(signal_Xa, fs);
end

% Combine DF and SC as features
X_features = [DF_Xa, SC_Xa]; % Feature matrix (DF and SC)

% 5-Fold Cross-Validation
cv = cvpartition(y, 'KFold', 5); % Create 5-fold cross-validation partition
accuracy = zeros(cv.NumTestSets, 1); % Initialize accuracy for each fold

for fold = 1:cv.NumTestSets
    % Split data into training and testing sets
    train_idx = training(cv, fold);
    test_idx = test(cv, fold);

    X_train = X_features(train_idx, :);
    y_train = y(train_idx);
    X_test = X_features(test_idx, :);
    y_test = y(test_idx);

    % Train Logistic Regression Model
    model = fitglm(X_train, y_train, 'Distribution', 'binomial', 'Link', 'logit');

    % Predict on test set
    y_pred = round(predict(model, X_test)); % Logistic regression outputs probabilities, round
        to 0 or 1

    % Compute accuracy
    accuracy(fold) = sum(y_pred == y_test) / length(y_test);
end

% Display overall performance
mean_accuracy = mean(accuracy) * 100;
fprintf('Mean Accuracy (5-Fold Cross-Validation): %.2f%%\n', mean_accuracy);

```

```
*****output result*****
```

```
Mean Accuracy (5-Fold Cross-Validation): 56.00%
```

– 5-Fold Cross-Validation Result:

- Mean accuracy for predicting AF recurrence using Dominant Frequency (DF) and Spectral Concentration (SC) is **56%**.
- An accuracy of **56%** means the classifier correctly predicts AF recurrence in 56% of the test samples on average across the 5 folds.
- This performance is slightly better than random guessing (**50%** for binary classification).
- However, the accuracy is relatively low, indicating that DF and SC **alone** may not be sufficient for reliable AF recurrence prediction.
- Additional features or more complex models might be necessary to improve prediction performance.

1.5 Repeat parts 2 and 4 by averaging, for each patient, the spectral parameters over all precordial leads (V1 to V6).

```

% Parameters
fs = 256; % Sampling frequency
leads = 7:12; % Precordial leads V1 to V6
num_patients = size(Xa_all, 3); % Total number of patients

% Initialize results
DF_results = zeros(num_patients, 1); % Averaged Dominant Frequency for each patient
SC_results = zeros(num_patients, 1); % Averaged Spectral Concentration for each patient

% Loop through all patients
for patient = 1:num_patients
    DF_leads = zeros(length(leads), 1); % DF for each lead
    SC_leads = zeros(length(leads), 1); % SC for each lead

    % Loop through all precordial leads (V1 to V6)
    for lead_idx = 1:length(leads)
        lead = leads(lead_idx);
        signal = squeeze(Xa_all(lead, :, patient)); % Extract signal for current lead and patient

        % Compute DF and SC for the current lead
        [DF_leads(lead_idx), SC_leads(lead_idx)] = compute_df_sc(signal, fs);
    end

    % Average DF and SC over all leads (V1 to V6)
    DF_results(patient) = mean(DF_leads);
    SC_results(patient) = mean(SC_leads);
end

% Display summary results
fprintf('Mean Dominant Frequency (DF): %.2f Hz\n', mean(DF_results));
fprintf('Mean Spectral Concentration (SC): %.2f %\n', mean(SC_results));

% Plot DF and SC distributions
figure;
subplot(2, 1, 1);
histogram(DF_results, 10, 'FaceColor', 'b');
title('Dominant Frequency (DF) Distribution (Averaged Over V1 to V6)');
xlabel('Frequency (Hz)'); ylabel('Count');

subplot(2, 1, 2);
histogram(SC_results, 10, 'FaceColor', 'r');
title('Spectral Concentration (SC) Distribution (Averaged Over V1 to V6)');
xlabel('SC (%)'); ylabel('Count');

```

```

*****output result*****
Mean Dominant Frequency (DF): 6.60 Hz
Mean Spectral Concentration (SC): 46.23 %

```

```

% Extract recurrence labels
load('C:\Users\praba\Documents\GitHub\UCA-M2-SEMESTER2\bio medical signal processing\lab assignments\Atrial_prabal_ghosh\indrecur.mat');
load('C:\Users\praba\Documents\GitHub\UCA-M2-SEMESTER2\bio medical signal processing\lab assignments\Atrial_prabal_ghosh\indnonrecur.mat');
y = zeros(num_patients, 1); % Initialize labels
y(indrecur) = 1; % Recurrence patients labeled as 1
y(indnonrecur) = 0; % Non-recurrence patients labeled as 0

% Combine DF and SC as features
X_features = [DF_results, SC_results]; % Feature matrix (averaged DF and SC)

```

```

% Custom 5-Fold Cross-Validation
k = 5; % Number of folds
indices = mod(1:num_patients, k) + 1; % Generate fold indices manually
accuracy = zeros(k, 1); % Initialize accuracy for each fold

for fold = 1:k
    % Split data into training and testing sets
    test_idx = (indices == fold);
    train_idx = ~test_idx;

    X_train = X_features(train_idx, :);
    y_train = y(train_idx);
    X_test = X_features(test_idx, :);
    y_test = y(test_idx);

    % Train Logistic Regression Model
    model = fitglm(X_train, y_train, 'Distribution', 'binomial', 'Link', 'logit');

    % Predict on test set
    y_pred = round(predict(model, X_test)); % Logistic regression outputs probabilities, round
        to 0 or 1

    % Compute accuracy
    accuracy(fold) = sum(y_pred == y_test) / length(y_test);
end

% Display overall performance
mean_accuracy = mean(accuracy) * 100;
fprintf('Mean Accuracy (5-Fold Cross-Validation): %.2f%%\n', mean_accuracy);

```

```
*****output result*****
```

```
Mean Accuracy (5-Fold Cross-Validation): 56.00%
```

- **Mean Dominant Frequency (DF): 6.60 Hz**
 - The mean DF of **6.60 Hz** lies within the typical atrial fibrillation (AF) band (3–9 Hz).
 - This suggests that the dominant atrial activity frequency is consistent with known AF characteristics.
- **Mean Spectral Concentration (SC): 46.23%**
 - The mean SC of **46.23%** indicates that nearly half of the power in the AF band is concentrated around the dominant frequency.
 - This suggests that the atrial signals are **moderately clean**, though some contamination from ventricular components or noise may still be present.
- **Mean Accuracy (5-Fold Cross-Validation): 56.00%**
 - The classifier achieved a mean accuracy of **56%** in predicting AF recurrence using the averaged DF and SC values across precordial leads (V1 to V6).
 - This performance is slightly better than random guessing (**50%**) but suggests that DF and SC alone may not be sufficient for reliable prediction.
- **Possible Reasons for Low Accuracy:**
 - **Limited Features:** DF and SC may not fully capture the complexity of AF recurrence. Incorporating additional features (e.g., time-domain metrics, heart rate variability) could enhance predictive performance.
 - **Simple Classifier:** Logistic regression is a basic model. More advanced machine learning methods (e.g., support vector machines, random forests, or neural networks) may yield better results.
- **Comparison with Single-Lead Analysis**
 - **Averaging Over V1 to V6:** Averaging DF and SC over all precordial leads (V1 to V6) provides a more robust and comprehensive representation of atrial activity compared to using a single lead (e.g., V1).
 - **Limited Accuracy Improvement:** Despite the broader representation, the improvement in classification accuracy is minimal.

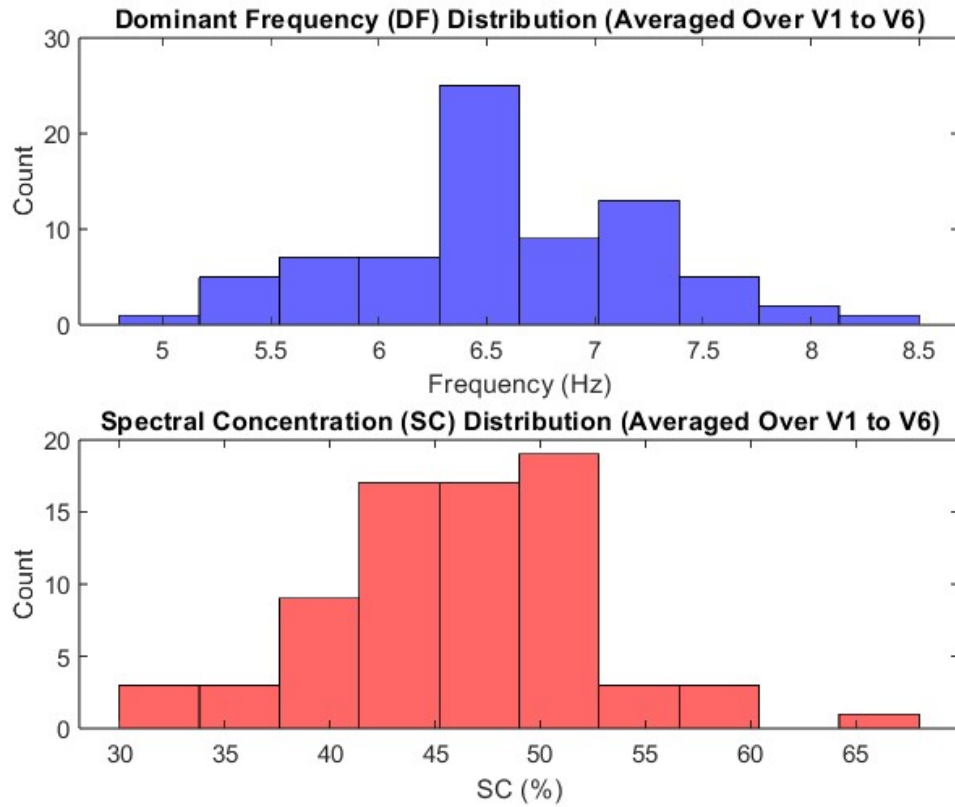


Fig. 8: PSD of original and Filtered ECG Signal

1.6 Conclude on the ability of spectral parameters DF and SC to predict AF recurrence in the given dataset.

– Conclusion on Predictive Ability of DF and SC:

- Using Dominant Frequency (DF) and Spectral Concentration (SC) alone yields a mean 5-fold cross-validation accuracy of only **56%**, only marginally above chance (50%).
- The substantial overlap in DF and SC distributions between patients with and without AF recurrence limits the classifier's discriminative power.
- These spectral markers, while physiologically relevant, do not capture all factors driving AF recurrence.
- To improve prediction performance, it is necessary to:
 - * Explore more advanced machine-learning models (e.g., random forests, support vector machines, deep neural networks).
- In summary, DF and SC are valuable for characterizing atrial activity but are *insufficient alone* for reliable AF recurrence prediction in the current dataset.